

Transformations of Functions

Name: _____

Pre-Calculus | Guided notes — we fill these in together

Vocabulary

Translation: a _____ of the graph — $f(x)+k$ moves it _____, and $f(x-h)$ moves it _____.

Reflection: a flip: $-f(x)$ flips over the _____, and $f(-x)$ flips over the _____.

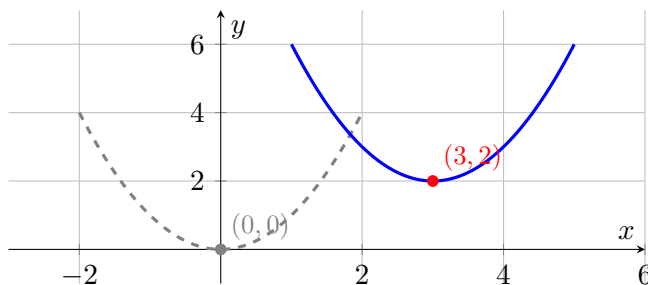
We do together. Describe how $g(x) = (x - 3)^2 + 2$ transforms the parent x^2 .

Step 1. Inside the square, $x - 3$ shifts the graph _____ (opposite the sign).

Step 2. The $+2$ outside shifts it _____.

Step 3. So the vertex moves from $(0, 0)$ to _____.

See it:



parent (dashed) $\rightarrow$ right 3, up 2 (solid).

You Try. Solve each. Show every step, then name the answer.

a) Describe $h(x) = -x^2$ from the parent.

b) Give the vertex of $y = (x + 1)^2 - 4$.

Summary (fill in): $f(x - h) + k$ shifts a graph _____; a leading $-f(x)$ _____ it over the x -axis.